



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Parallel Computing–I part–Thursday, July 04th, 2024

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

- A. Please define how efficiency can be computed when a PRAM model is used (1)

- B. Please list the concurrent read the prefix sum algorithm has when a PRAM model is used. (1)

- C. Please show an example of the PRAM model solving CW with a common CW approach. (1)

- D. Is the absolute serial time fixed with the Gustafson model? What about the parallel part? (1)

Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Please describe the key characteristics of data-parallel program execution with SIMD instructions. (2)

[illegible]

- B. Explain how multithreading approaches the memory stall problem. (1)

- C. Please compare Coarse-grain multithreading with Simultaneous multithreading. (1)

Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

A. Please describe what is meant by **GPU "SIMT"**. (1)

B. Please describe the key characteristics of the **CUDA synchronization** constructs. (1)

C. Please describe how each sub-core runs the next instruction for the CUDA threads in the warp. What is meant by divergence in this context? (2)

Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

- A. Please briefly describe when uncoalesced memory access happened. (1) **This question can be skipped in case of a passed challenge.**

- B. Please briefly describe how DRAM Bursting Timing is. (1)

- C. Please briefly describe with an example how the "Corner turning" technique works. Why is improving the kernel performance and in which case? (2)
